

The Scenario: "Our retail chain is deploying an autonomous AI Agent to manage inventory for 500 stores. The agent is responsible for analyzing local sales trends, checking warehouse stock, and drafting purchase orders. However, in pilot testing, the agent occasionally hallucinated sales figures, ignored warehouse constraints, and once tried to order 10,000 units of a product that was being discontinued."

The Core Task:

"Explain how you would apply Prompt, Context, and Harness Engineering to move this agent from an unreliable pilot to a production-ready system. Specifically, outline the 'Safety & Reliability' architecture you would build around the LLM."

Prompt Engineering define on how the AI think which is in this situation AI have hallucinated sales figure which is it skipped the important task to check the warehouse constraint and generated inaccurate demand data. Without proper guidance, the LLM tend to relied on their own reasoning, which led to inaccurate and unsafe outputs.

Prompt engineering improves the AI reasoning by providing clear and specific instructions. The system prompt should define the AI's role, require it to use only verified data, follow a structured workflow such as forecast demand, check stock, verify constraints, and lastly draft purchase order, and escalate any unusual or unsure decision for human review instead of making their own decision. Few example can be included to show the correct response for them when products are discontinued or when order quantities are unusually high.

For the context engineering, this shows the AI hallucinated sales figure and ignore important business information because it lack of access to real data and also updated data. Instead of using real sales records and inventory information, it relied on the internal knowledge, resulting in incorrect forecast and purchase decision.

To solve this issue, it must provide AI with reliable sources and truth of each data that can be retrieved from the company database and APIs. This includes all the information which is the sale trends, warehouse stock updated level, product availability, and product status which can be seen as active or discontinued By giving the AI with accurate context, it no longer to guess all the information which is can reduce hallucinations in term of guessing the data and improving the accuracy of inventory recommendations.

For the harness engineering, even with a good prompts and accurate data, the AI could still generate unsafe output, such as ordering 10000 units of a discontinued product or

exceeding warehouse capacity. This allowing AI to execute these decisions directly could lead to major financial losses and operational problem

Harness engineering acts as the final safety layer by validating every AI generated purchase order before executed. This can determine the business rule automatically can reject order for discontinued product, block quantities that exceed warehouse capacity or expected demands, and also require human for approval of any unusual or high value purchases. In addition, the AI is only allowed to draft purchase order, while the final decision and execution only remain outside the AI's control which this can ensure there are no unsafe action and can always be monitored.